PHYS1408 Practice Exam 2-3 — Answer Key

Worked solutions

All answers were checked numerically with `_scripts/verify_2_3.py`. $g = 9.8$ m/s^2 throughout.

What this exam covers:

- **The four question types not on Exams 2-1 or 2-2:** free fall (Q4), work and the definition of energy (Q2, Q3), the cue ball and 8-ball (Q9), and a non-spinning disk dropped on a disk (Q14).
- **Eleven twists on types you've already seen:** each asks a *different* unknown or adds a new condition.

Q	Answer	Value	Topic (source)	New or variation
1	**d**	22.7 m/s → 21	Banked v_{max}, round **down** (Ch 6 notes p. 3/7–5/7)	variation: max trap
2	**e**	—	Definition of energy (Ch 7/8 notes p. 1/18)	**new**
3	**c**	520 J	Work $W = Fd\cos\phi$ (Ch 7/8 notes p. 1/18)	**new**
4	**a**	29.7 m/s	Free fall, "coyote" (Ch 7/8 notes p. 1/18)	**new**
5	**d**	4 crossings	Halfpipe, several passes (Ch 7/8 notes p. 4/18–5/18)	variation
6	**e**	4.85 m	Rough incline up, stopping distance (Ch 7/8 notes p. 9/18)	variation: distance, not time
7	**b**	6.32 m/s	Spring → kinetic energy (Ch 7/8 notes p. 11/18)	variation
8	**b**	4900 N	Loop, N at the top (Ch 7/8 notes p. 12/18–13/18)	variation
9	**c**	0	Cue ball hits 8-ball (Ch 9 notes p. 4/5)	**new**

Q	Answer	Value	Topic (source)	New or variation
10	**e**	-0.67 m/s	Elastic, both moving the same way (Ch 9 notes p. 3/5)	variation
11	**b**	0.20 m	Ballistic pendulum, find h (Ch 9 notes p. 5/5)	variation
12	**c**	16.8 N	Pulley with mass, tension (Ch 10 notes p. 4/8)	variation
13	**a**	3.83 m/s	Rolling hoop (Ch 10 notes p. 7/8–8/8)	variation
14	**d**	3.27 rad/s	Disk dropped on a disk (Ch 11 notes p. 1/8)	**new**
15	**a**	0.31 rad/s	Child runs **against** the rotation (Ch 11 notes p. 4/8)	variation

1. Banked curve, maximum speed — d) 21 m/s

At v_{max} the car tends to slide **up** the bank, so friction points **down** the bank (Ch 6 notes p. 4/7):

$$v_{max} = \sqrt{gr\,\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}} = \sqrt{9.8 \times 78 \times \frac{0.25 + 0.364}{1 - 0.25 \times 0.364}} = \sqrt{764.4 \times 0.6754} = \sqrt{516.3} = 22.7 \text{ m/s}.$$

Answer-choice trap. 22.7 m/s is a **maximum**. At 23 m/s the car slides up the bank, so e) 23 fails even though it's the closest number. The largest choice that works is **d) 21 m/s**.

Distractors:

- a) 9 is $v_{min} = 8.9$.
- b) 14 is the flat curve, $\sqrt{\mu_s g r} = 13.8$.
- c) 17 is the frictionless bank, $\sqrt{gr\tan\theta} = 16.7$.

2. Definition of energy — e) the ability to do work

The professor's notes open Ch 7/8 with "**Energy is the ability to do work.**" Answer with his wording.

Every distractor is his definition of something else:

- a) is **power** ($P = W/t$).
- b) is wrong twice: energy is a scalar, measured in joules.
- c) is **momentum** ($p = mv$).
- d) is the **moment of inertia** ("I is resistance to rotation").

3. Work done by a rope at an angle — c) 520 J

$W = Fd\cos\phi = \vec{F} \cdot \vec{d}$. Only the component of the force along the motion does work (Ch 7/8 notes p. 1/18):

$$W = 50 \times 12 \times \cos 30^\circ = 600 \times 0.866 = 520 \text{ J}.$$

(ϕ = angle between the force and the direction of motion.)

Distractors:

- a) 0 is for a force perpendicular to the motion.
- b) 300 uses $\sin\phi$, the vertical part, which does no work.
- d) 600 ignores the angle.
- e) 1200 doubles it.

4. Coyote falls off a cliff — a) 29.7 m/s

No friction, so $E_i = E_f$ (Ch 7/8 notes p. 1/18). Put $h = 0$ at the ground; it starts from rest.

Energy terms:

$$E_i = mgh$$

$$E_f = \tfrac{1}{2}mv^2$$

Solve for v:

$$mgh = \tfrac{1}{2}mv^2$$

$$v^2 = 2gh$$

$$v = \sqrt{2gh}$$

Plug in:

$$v = \sqrt{2(9.8)(45)} = \sqrt{882} = 29.7 \text{ m/s}$$

Distractors:

- b) 42.0 doubles the height.
- c) 59.4 doubles the answer.
- d) 441 is gh with no root.
- e) 882 is $2gh$ with no root.

5. Halfpipe, several passes — d) 4

Each crossing of the rough section removes $E_{lost} = \mu mgL$ (Ch 7/8 notes p. 5/18), so the height she can reach drops by $\mu L = 0.10 \times 4.0 = 0.40$ m per crossing:

Crossing	Height reached afterward
start	1.8 m
1	1.4 m
2	1.0 m
3	0.6 m
4	0.2 m
5	needs 0.40 m, only has 0.20 m, so she stops partway

On the fifth trip she has $mg(0.2)$ of energy left. Setting $mgh_{left} = \mu mg\,x$ gives $x = \dfrac{h_{left}}{\mu} = \dfrac{0.2}{0.10} = 2.0$ m, so she stops 2.0 m into the patch, halfway across. She crossed the **entire** section **4** times.

Distractor e) 5 rounds $1.8/0.40 = 4.5$ up.

6. Rough incline, sliding up, stopping distance — e) 4.85 m

$E_i = E_f + E_{lost}$ (Ch 7/8 notes p. 9/18). Put $h = 0$ at the start; it rises $\Delta x \sin\theta$ and ends at rest. $N = mg\cos\theta$ on the incline.

Energy terms:

$$E_i = \tfrac{1}{2}mv_i^2$$

$$E_f = mg\Delta x \sin\theta$$

$$E_{lost} = \mu_k mg\cos\theta\,\Delta x$$

Solve for Δx:

$$\tfrac{1}{2}mv_i^2 = mg\Delta x\sin\theta + \mu_k mg\cos\theta\,\Delta x$$

$$\tfrac{1}{2}v_i^2 = g\Delta x(\sin\theta + \mu_k\cos\theta)$$

$$\Delta x = \frac{v_i^2}{2g(\sin\theta + \mu_k\cos\theta)}$$

Plug in:

$$\Delta x = \frac{8.0^2}{2(9.8)(0.5 + 0.20\times 0.866)} = \frac{64}{19.6(0.673)} = 4.85 \text{ m}$$

The 2.0 kg mass cancels. If you're also asked for the **time** (as on Exam 2-1 #4), use $\Delta x = \frac{v_i + 0}{2} t$, so $t = \frac{2\Delta x}{v_i} = 1.21$ s.

Distractors:

- a) 0.61 forgets to square v_i.
- b) 2.42 has an extra 2.
- c) 3.27 is $v_i^2/2g$, the height if it were thrown straight up.
- d) 4.66 uses friction μmg instead of $\mu mg \cos\theta$.

7. Horizontal spring launcher — b) 6.32 m/s

No friction, so $E_i = E_f$ (Ch 7/8 notes p. 11/18). The floor is level, so gravity does nothing. Spring PE becomes KE.

Energy terms:

$$E_i = \tfrac{1}{2}kx^2$$

$$E_f = \tfrac{1}{2}mv^2$$

Solve for v:

$$\tfrac{1}{2}kx^2 = \tfrac{1}{2}mv^2$$

$$v^2 = \frac{kx^2}{m}$$

$$v = \sqrt{\frac{kx^2}{m}} = x\sqrt{\frac{k}{m}}$$

Plug in:

$$v = 0.10\sqrt{\frac{800}{0.20}} = 0.10(63.2) = 6.32 \text{ m/s}$$

Distractors:

- a) 4.47 drops one $\frac{1}{2}$.
- c) 8.94 drops the other $\frac{1}{2}$.
- d) 20.0 forgets to square x.
- e) 40.0 forgets the square root.

8. Loop, normal force at the top — b) 4900 N

Step 1: Speed at the top (energy, $E_i = E_f$). Put $h = 0$ at the bottom of the loop; the car starts at rest at h, and the top of the loop is at $2R$.

$$E_i = mgh$$

$$E_f = mg(2R) + \tfrac{1}{2}mv_{top}^2$$

$$mgh = mg(2R) + \tfrac{1}{2}mv_{top}^2$$

$$v_{top}^2 = 2g(h - 2R)$$

$$v_{top}^2 = 2(9.8)(30 - 20) = 196 \text{ m}^2/\text{s}^2$$

Step 2: Forces at the top. N and mg both point **down**, toward the center (Ch 7/8 notes p. 12/18):

$$N + mg = \frac{mv_{top}^2}{R}$$

$$N = \frac{mv_{top}^2}{R} - mg$$

Step 3: Plug in.

$$N = \frac{500(196)}{10} - 500(9.8) = 9800 - 4900 = 4900 \text{ N}$$

That's exactly mg: the track pushes down on the car with a force equal to the car's weight. Launching from the minimum height $h = \frac{5}{2}R = 25$ m would give $N = 0$.

Distractors:

- a) 0 is the "barely" case.
- c) 9800 forgets to subtract mg.
- d) 14,700 adds mg (the bottom-of-loop sign).
- e) 24,500 uses the full 30 m drop.

9. Cue ball hits the 8-ball — c) 0

For equal masses with the 8-ball at rest (Ch 9 notes p. 4/5):

- Momentum: $v_{1i} = v_{1f} + v_{2f}$.
- Elastic: $v_{1i} + v_{1f} = v_{2f}$.

Substituting the second into the first gives $v_{1f} = 0$: "cue is still," and the 8-ball leaves at 3.0 m/s. The notecard formula agrees:

$$v_{1f} = \frac{(m-m)v_{1i} + 2m(0)}{2m} = 0.$$

Distractors:

- d) +1.5 is the "they stick" (inelastic) answer.
- e) +3.0 is the 8-ball's velocity, not the cue ball's.
- a) −3.0 is a full bounce-back, as off a wall.

10. Elastic collision, both moving right — e) −0.67 m/s

Using the notecard formulas with right as positive (Ch 9 notes p. 3/5): $m_1 = 2.0$, $v_{1i} = +6.0$, $m_2 = 4.0$, $v_{2i} = +1.0$.

$$v_{1f} = \frac{(m_1 - m_2)v_{1i} + 2m_2 v_{2i}}{m_1 + m_2} = \frac{(-2)(6.0) + 2(4.0)(1.0)}{6.0} = \frac{-12 + 8}{6} = -0.67 \text{ m/s}.$$

Check:

- $v_{2f} = v_{1i} + v_{1f} - v_{2i} = 6 - 0.67 - 1 = +4.33$ m/s.
- Momentum: $12 + 4 = 16$ before; $-1.33 + 17.33 = 16$ after. (checks)
- KE: $36 + 2 = 38$ J before; $0.44 + 37.56 = 38$ J after. (checks)

The 2 kg ball bounces back to the left.

Distractors:

- a) +4.33 is the 4 kg ball.
- b) +2.67 is the stick-together answer.
- c) +0.67 drops the sign.
- d) 0 assumes the cue-ball result, which only works for equal masses with one at rest.

11. Ballistic pendulum, swing height — b) 0.20 m

Two stages, two laws (Ch 9 notes p. 5/5).

1. **Collision** (momentum only): $mv = (M+m)V$, so $V = \dfrac{mv}{M+m} = \dfrac{0.010 \times 400}{2.010} = 1.99$ m/s.
2. **Swing** (energy): $\frac{1}{2}(M+m)V^2 = (M+m)gh$, so $h = \dfrac{V^2}{2g} = \dfrac{1.99^2}{19.6} = 0.20$ m.

Distractors:

- a) 0.10 forgets to square V.
- c) 0.40 forgets the 2.
- d) 1.99 is V, not a height.
- e) 40.6 m uses energy conservation through the collision, $\frac{1}{2}mv^2 = (M+m)gh$. **Never** use energy through the collision: here 99.5% of the bullet's KE is lost in it.

12. Tension over a ring pulley — c) 16.8 N

Torque on the pulley comes from the tension (Ch 10 notes p. 4/8): $RT = I\alpha = MR^2(a/R)$, so $T = Ma$. For the hanging mass, $mg - T = ma$. Combining:

$$a = \frac{mg}{m+M} = \frac{3.0 \times 9.8}{7.0} = 4.2 \text{ m/s}^2, \qquad T = Ma = 4.0 \times 4.2 = 16.8 \text{ N}.$$

Check from the hanging mass: $T = m(g-a) = 3.0(9.8 - 4.2) = 16.8$ N. (checks)

Distractors:

- a) 0 treats the pulley as massless with nothing on the other side.
- b) 11.8 uses the disk formula, $I = \frac{1}{2}MR^2$.
- d) 29.4 is mg (true only if nothing moves).
- e) 42.0 is $m(g+a)$, a sign error.

13. Rolling hoop — a) 3.83 m/s

$E_i = E_f$ (Ch 10 notes p. 7/8). Put $h = 0$ at the bottom. Rolling means $R\omega = v$, and a hoop has $I = MR^2$.

Energy terms:

$$E_i = mgh$$

$$E_f = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I\omega^2$$

Solve for v:

$$mgh = \tfrac{1}{2}mv^2 + \tfrac{1}{2}\left(mR^2\right)\frac{v^2}{R^2} = \tfrac{1}{2}mv^2 + \tfrac{1}{2}mv^2 = mv^2$$

$$v = \sqrt{gh}$$

Plug in:

$$v = \sqrt{9.8 \times 1.5} = 3.83 \text{ m/s}$$

From his table: ring $\sqrt{1.0\,gh}$, disk $\sqrt{1.33\,gh}$, sphere $\sqrt{1.43\,gh}$, slide $\sqrt{2gh}$.

Distractors:

- b) 4.43 is a disk.
- c) 4.58 is a sphere.
- d) 5.42 is sliding.
- e) 7.67 is $2\sqrt{gh}$.

The hoop is slowest because half its energy goes into spinning.

14. Non-spinning disk dropped on a spinning disk — d) 3.27 rad/s

$L_i = L_f$, and the top disk starts with $\omega_{2i} = 0$ (Ch 11 notes p. 1/8):

$$I_1 = \tfrac{1}{2}(200)(1.5)^2 = 225 \text{ kg} \cdot \text{m}^2, \qquad I_2 = \tfrac{1}{2}(100)(1.0)^2 = 50 \text{ kg} \cdot \text{m}^2.$$

$$\omega_f = \frac{I_1\omega_{1i}}{I_1 + I_2} = \frac{225 \times 4.0}{275} = 3.27 \text{ rad/s}.$$

Distractors:

- a) 2.00 halves ω.
- b) 2.67 uses the mass ratio 200/300 instead of I.
- c) 2.77 treats the dropped disk as a point mass, mr^2.
- e) 4.00 is no change.

15. Child runs against the rotation — a) 0.31 rad/s

The child's angular momentum about the axle is $L = mvR$ (Ch 11 notes p. 4/8). Because the child runs **opposite** to the rim, it is **negative** (counter-rotating, as in Ch 11 notes p. 3/8):

$$\omega_f = \frac{\tfrac{1}{2}MR^2\omega_i - mvR}{\tfrac{1}{2}MR^2 + mR^2} = \frac{400(1.0) - 30(4.0)(2.0)}{400 + 30(2.0)^2} = \frac{400 - 240}{520} = 0.31 \text{ rad/s}.$$

The merry-go-round slows down but keeps turning in its original direction.

Distractors:

- b) 0.40 forgets to add the child's mR^2.
- c) 0.77 ignores the child's running.
- d) 1.23 uses + instead of −.
- e) 1.60 makes both mistakes.